GreenMind: Deep Reinforcement Learning for Sustainable Buildings

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Background

A smart building controller is responsible for managing the HVAC system in a building. The controller looks at real-time sensor data. The temperature, inside the building the level of carbon dioxide and how many people're present. And has to decide how to run fans adjust dampers and control chilled-water valves at every moment.

The environment is unpredictable. The weather outside changes, without warning. People. Go at random times. Even the HVAC system doesn't always respond the way it should. The controller has to handle three things at once: keeping the space comfortable saving energy and maintaining air quality. It's a job because all three goals pull in different directions.

- Minimize energy consumption. Minimizing energy consumption reduces cost to the building operator.
- Maintain comfort by keeping the indoor temperature between 21°C and 24°C.
- Keep air quality good by ensuring CO_2 stays below 1000 parts, per million.
- Avoid safety violations by making sure the temperature never goes above 32 degrees Celsius or drops, below 15 degrees Celsius.

The agent needs to learn a policy that increases the total reward over time. This means keeping the building at a temperature ensuring clean air and using as little energy, as possible. All of these goals should be achieved together without allowing conditions to become unsafe. An MDP is described as:

$$M = (S, A, T, R, \gamma)$$

1. State Space S

This shows what is happening with the building at each moment in time. The condition is made up of three things: the temperature Cold / Comfort / Hot) the level of CO_2 (Normal / High) and whether people are there (Occupied / Unoccupied).

S1 : The place where the temperature inside is just right (21°C to 24°C) the air has an amount of CO_2 (1000 ppm or less) and people are, inside the building.

S2 : The state where the indoor temperature has risen above 24°C (overheated), CO_2 is still normal, and the building is occupied. The space is thermally uncomfortable and needs cooling.

S3 : The state where the indoor temperature has dropped below 21°C (overcooled), CO_2 is normal, and the building is occupied. The space is too cold and needs to be warmed up.

S4 : The state where the indoor temperature is within the comfort zone, but CO_2 has exceeded 1000 ppm (high), and the building is occupied. The temperature is fine but the air quality is poor — the space feels stuffy.

S5 : The state where the indoor temperature is above 24°C and CO_2 exceeds 1000 ppm, with the building occupied. This is the critical state — both thermal comfort and air quality are violated at the same time.

S6 : The state where the building is unoccupied. Temperature and CO_2 values do not matter for occupant comfort in this state, so all temperature and CO_2 combinations collapse into a single unoccupied state.

S7 : The terminal state reached when the indoor temperature violates the safety boundary — either falling below 15°C or rising above 32°C. The episode ends immediately in failure.

S8 : The terminal state reached when the agent has successfully sustained ideal indoor conditions (S1) for long enough that the operating day is declared complete. The episode ends in success with a reward of +15.

2. Action Space

At S1 — Comfort Zone (Ideal State, Occupied):

The system is running fine. The agent decides how to run the HVAC. Here the agent aims to achieve the Green Day state for the building (→ S8).

- **A1:** Maintain Mode — Keep current HVAC at low-moderate power.
- **A2:** Eco Mode — Reduce HVAC power significantly to save energy.
- **A3:** Boost Ventilation — Increase fresh air intake preemptively.

At S2 — Overheated (Hot, Normal CO_2, Occupied):

The room is too hot. The agent needs to cool the room.

- **B1:**Full Cooling – Chilled-water flow, high fan speed.
- **B2:**Moderate, Ventilation, Cooling is balanced and fresh air is present, Ventilation is balanced too.
- **B3:** Ventilation Boost Only – A way to increase airflow without extra chilled water.

At S3 — Overcooled (Cold, Normal CO_2, Occupied):

The room is too cold. The agent must warm the building up.

- **C1:** Reduce cooling by lowering the chilled-water flow and slowing down the fan speed, Let the room warm up naturally.
- **C2:** Activate Heating Mode – The Heating Mode will engage heat recovery and the reheat coil to actively warm the space.
- **C3:** Wait – Take no action; let the room return to balance on its own. All devices remain in their positions.

At S4 — Stuffy (Comfort Zone, High CO_2, Occupied):

Temperature is fine but air quality is poor. The agent must ventilate.

- **D1:** Boost Fresh Air — Increase damper opening and fan speed to dilute CO_2.

- **D2:** Full Ventilation — Maximum damper and fan to aggressively clear CO_2.
- **D3:** Do Nothing Mode — Keep HVAC as-is; hope CO_2 naturally dissipates. Lazy / passive action.

At S5 — Critical (Hot, High CO_2, Occupied):

Both temperature and air quality are bad. The agent must act decisively.

- **E1:** Full Cooling + Full Ventilation — Maximum combined response.
- **E2:** Full Cooling Only — Prioritise temperature; ignore CO_2 for now. .
- **E3:** Full Ventilation Only — Prioritise air quality; hope fresh air also cools slightly.

At S6 — Unoccupied Building:

No one is inside. The agent decides how much energy to use in the empty building.

F1: Standby Mode — Minimal HVAC to maintain safe temperature range.
F2: Full Shutdown — Turn off HVAC completely. Zero energy, but risk of extreme temperature.
F3: Pre-Condition — Run HVAC at moderate power in anticipation of occupants returning.

At S7 — Safety Violation (Negative Terminal):

The state ends in failure. The building has reached a dangerously extreme temperature. The terminal penalty of −15 is incurred.

At S8 — Green Day Achieved (Positive Terminal):

The building has successfully sustained ideal indoor conditions for long enough that the operating day is declared complete. Terminal reward of +15 is received upon entering this state.

4. Outcomes - Transitions probabilities and Rewards

Current State	Action	Probability	Reward
S1 Comfort	A1 Hold Steady	0.15	+15
S1 Comfort	A1 Hold Steady	0.55	−1
S1 Comfort	A1 Hold Steady	0.15	−3
S1 Comfort	A1 Hold Steady	0.10	−3
S1 Comfort	A1 Hold Steady	0.05	−1
S1 Comfort	A2 Eco Setting	0.10	+15
S1 Comfort	A2 Eco Setting	0.35	0
S1 Comfort	A2 Eco Setting	0.30	−2
S1 Comfort	A2 Eco Setting	0.15	−2
S1 Comfort	A2 Eco Setting	0.10	0
S1 Comfort	A3 Ventilation Boost	0.10	+15
S1 Comfort	A3 Ventilation Boost	0.65	−2
S1 Comfort	A3 Ventilation Boost	0.15	−4
S1 Comfort	A3 Ventilation Boost	0.10	−2
S2 High Temp.	B1 Set Cooling	0.75	−3
S2 High Temp.	B1 Set Cooling	0.15	−5
S2 High Temp.	B1 Set Cooling	0.05	−5
S2 High Temp.	B1 Set Cooling	0.05	−7
S2 High Temp.	B2 Mode Cooling and Ventilation	0.60	−2
S2 High Temp.	B2 Mode Cooling and Ventilation	0.30	−4
S2 High Temp.	B2 Mode Cooling and Ventilation	0.05	−4
S2 High Temp.	B2 Mode Cooling and Ventilation	0.05	−2
S2 High Temp.	B3 Vent Only	0.30	−2

Current State	Action	Probability	Reward
S2 High Temp.	B3 Vent Only	0.60	−4
S2 High Temp.	B3 Vent Only	0.10	−15
S3 Cold Temp.	C1 Less Cooling	0.70	−1
S3 Cold Temp.	C1 Less Cooling	0.20	−3
S3 Cold Temp.	C1 Less Cooling	0.05	−3
S3 Cold Temp.	C1 Less Cooling	0.05	−1
S3 Cold Temp.	C2 Set Heating	0.80	−2
S3 Cold Temp.	C2 Set Heating	0.15	−4
S3 Cold Temp.	C2 Set Heating	0.05	−4
S3 Cold Temp.	C3 Wait	0.40	0
S3 Cold Temp.	C3 Wait	0.45	−2
S3 Cold Temp.	C3 Wait	0.10	−15
S3 Cold Temp.	C3 Wait	0.05	0
S4 Stuffy	D1 Boost Air	0.70	−2
S4 Stuffy	D1 Boost Air	0.20	−4
S4 Stuffy	D1 Boost Air	0.05	−4
S4 Stuffy	D1 Boost Air	0.05	−2
S4 Stuffy	D2 Set Ventilation	0.80	−3
S4 Stuffy	D2 Set Ventilation	0.10	−5
S4 Stuffy	D2 Set Ventilation	0.05	−5
S4 Stuffy	D2 Set Ventilation	0.05	−3
S4 Stuffy	D3 Do Nothing	0.30	−1
S4 Stuffy	D3 Do Nothing	0.50	−3
S4 Stuffy	D3 Do Nothing	0.15	−5
S4 Stuffy	D3 Do Nothing	0.05	−1

Current State	Action	Probability	Reward
S5 Critical	E1 Set Cooling and Ventilation	0.60	−4
S5 Critical	E1 Set Cooling and Ventilation	0.20	−6
S5 Critical	E1 Set Cooling and Ventilation	0.10	−6
S5 Critical	E1 Set Cooling and Ventilation	0.05	−8
S5 Critical	E1 Set Cooling and Ventilation	0.05	−4
S5 Critical	E2 Set Cooling	0.55	−5
S5 Critical	E2 Set Cooling	0.20	−7
S5 Critical	E2 Set Cooling	0.15	−15
S5 Critical	E2 Set Cooling	0.10	−3
S5 Critical	E3 Full Vent Only	0.40	−4
S5 Critical	E3 Set Ventilation	0.30	−6
S5 Critical	E3 Set Ventilation	0.20	−15
S5 Critical	E3 Set Ventilation	0.10	−2
S6 Empty room	F1 On Hold	0.80	−1
S6 Empty room	F1 On Hold	0.15	−1
S6 Empty room	F1 On Hold	0.05	−3
S6 Empty room	F2 Switch Off	0.70	0
S6 Empty room	F2 Switch Off	0.15	−2
S6 Empty room	F2 Switch Off	0.10	−2
S6 Empty room	F2 Switch Off	0.05	−15
S6 Empty room	F3 Set Conditioned air	0.75	−2
S6 Empty room	F3 Set Conditioned air	0.15	−2
S6 Empty room	F3 Set Conditioned air	0.05	−4
S6 Empty room	F3 Set Conditioned air	0.05	−4

Current State	Action	Probability	Reward
S7 Safety Viol.	—	1.00	0 (terminal, bad)
S8 Green Day	—	**1.00**	**0 (terminal, success)**

5. Discount Factor γ

For GreenMind we set γ = 0.95.

- A high γ is appropriate because energy management is a long-horizon problem. The benefit of pre-conditioning the building (Action F3 at State S6) is only realised several steps later when occupants arrive in S1 instead of S2.

- A small γ would make the agent short-sighted — it would prefer immediate energy savings even at the cost of future discomfort.

6. Policy

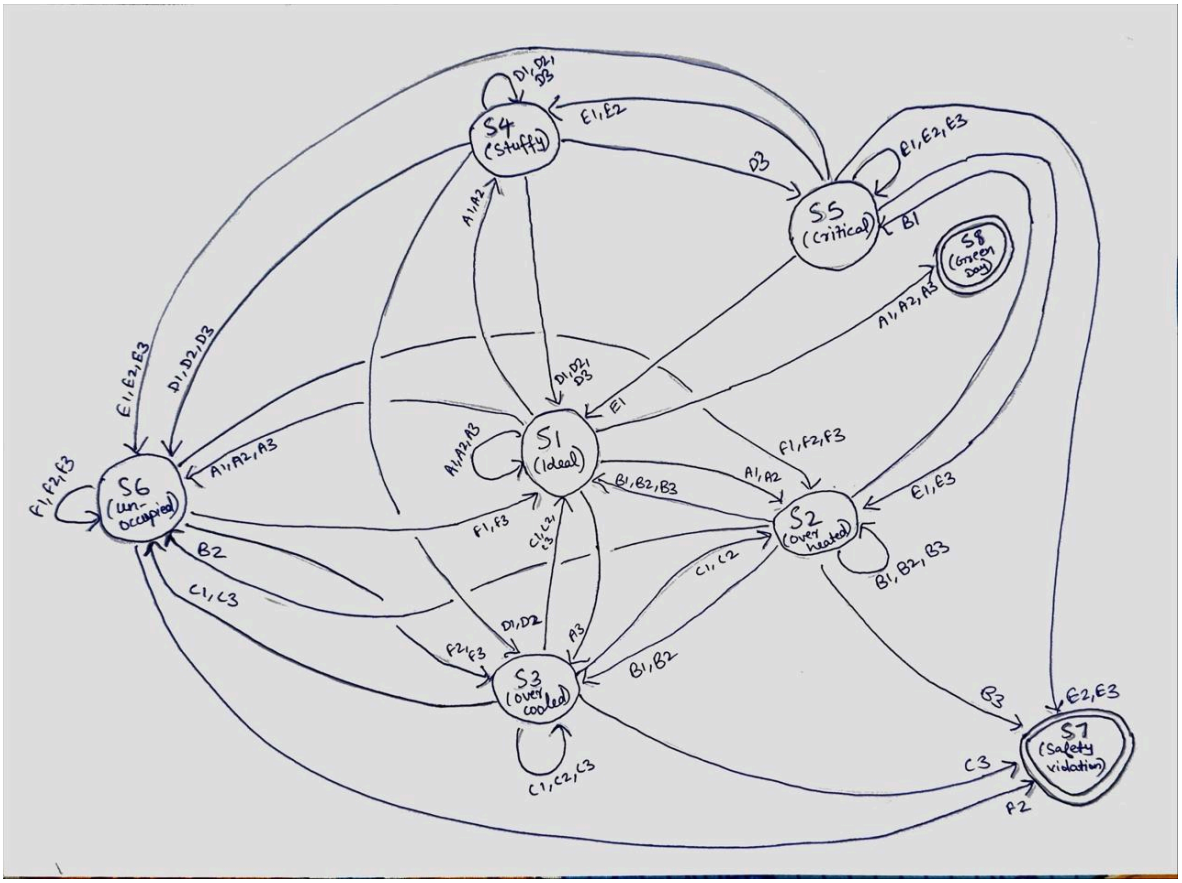
To specify what HVAC control command the agent must issue at every building condition it encounters, we look for the best policy.

Since the GreenMind problem has 6 non-terminal states, the policy must define a decision for each one. At S1 (Comfort Zone, Normal CO_2, Occupied), the policy selects from {A1, A2 or A3}. At S2 (Overheated), it selects from {B1, B2 or B3} and so on. For a deterministic policy, one fixed action is assigned per state.

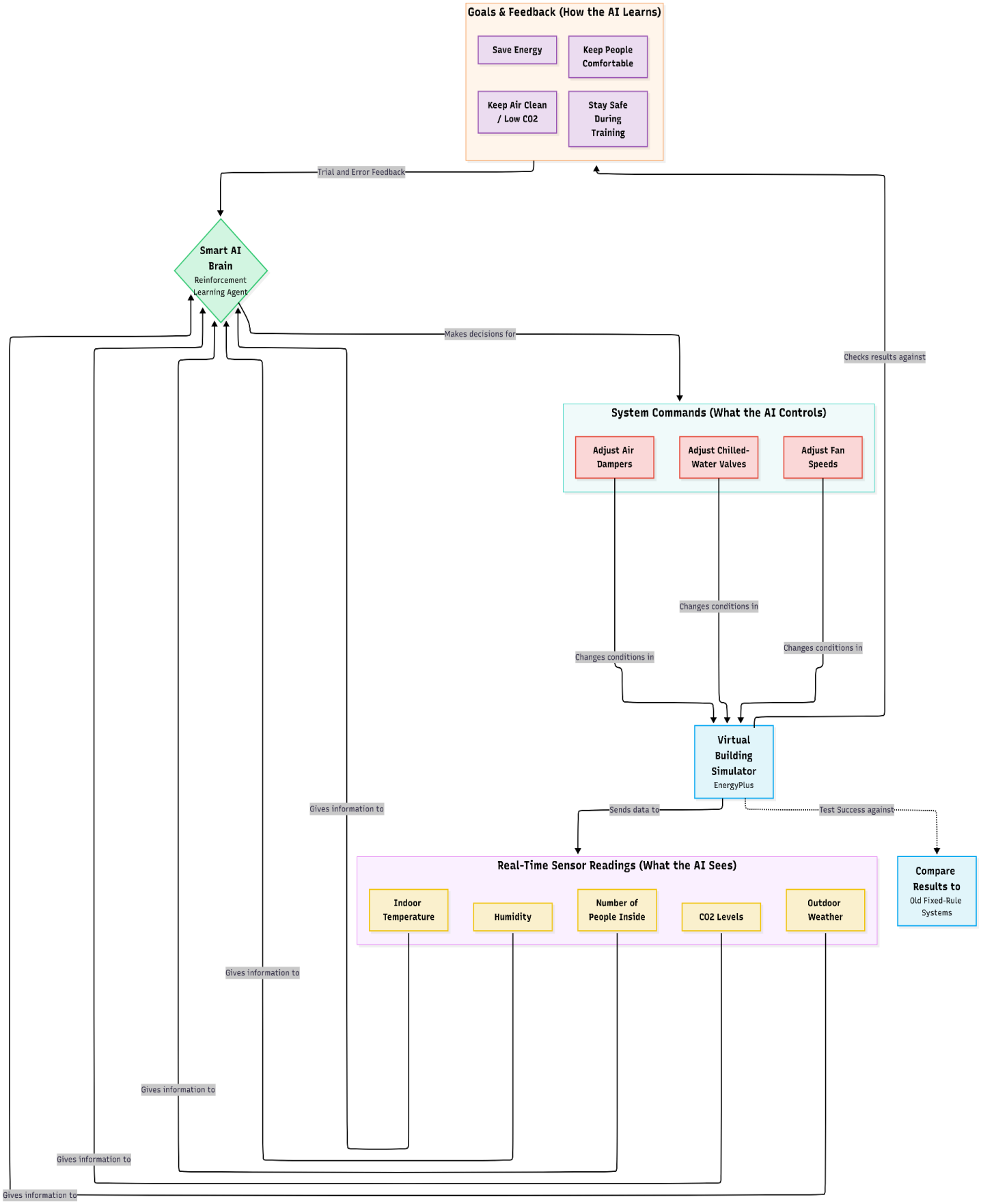
7. Objective

The objective of GreenMind is to find the optimal policy **π*** that maximises the expected discounted cumulative reward collected by the HVAC agent over one full operating day, where γ = 0.95 and T is the episode length.

This is solved using Policy Iteration — a recursive process of Policy Evaluation and Policy Improvement repeated until convergence. Policy Evaluation computes the value of every building state under the current policy using the Bellman expectation equation.



8. High - Level Solution Diagram



9. DATASET

1. **Building Data Genome Project 2 (BDG2)**
   An open dataset containing hourly interval energy meter data (electricity, heating, cooling) and weather data from over 1,600 non-residential buildings.
   https://github.com/buds-lab/building-data-genome-project-2

2. **Occupancy Detection from Environmental Data**
   Experimental time-series data from an office room measuring indoor temperature, humidity, light, and CO2 levels, paired with ground-truth binary occupancy labels.
   https://archive.ics.uci.edu/ml/datasets/occupancy+detection

3. **Sinergym (Simulation Framework)**
   A Gymnasium-based Python interface for the EnergyPlus physics engine used to create custom interactive building simulation environments for reinforcement learning agents.
   https://github.com/ugr-sail/sinergym

10. Evaluation Baselines

To rigorously check how well the model-free deep reinforcement learning policy works the agent's performance will be compared to two main control strategies. The first one is Rule-Based Control. This is the industry standard. Rule-Based Control uses set schedules, fixed temperature limits and simple rules to run heating, ventilation and air conditioning systems. Comparing the reinforcement learning agent to Rule-Based Control will show how well the agent can cut down on wasted energy and adjust to changing building conditions when areas are empty.

The second baseline is Model Predictive Control (MPC). MPC is a method that uses optimization. It uses math models of how buildings heat and cool. It also uses weather forecasts. The goal is to predict and reduce energy use in the future. Testing the policy against MPC will show how well the model-free reinforcement learning approach works. It will show if it can match or do better than perfect control. This does not need models built specifically for each building.

11. Key Performance Metrics

The evaluation framework measures how well the agent optimizes its reward function. The primary metric for system efficiency is energy consumption, which tracks the amount of power used by the continuous operations of dampers, fans and chilled-water control valves, during the evaluation period. This gives a measurement of the policy’s energy optimization capabilities.

To measure how well people feel in the building the project will watch two things: thermal comfort and indoor air quality. Thermal comfort will be checked by looking at how the indoor temperature goes outside the comfortable range. It will also use comfort measures to make sure the environment stays pleasant. This will be done when the weather outside changes and when the number of people, in the building changes. Indoor Air Quality or IAQ will be monitored using carbon dioxide levels. CO2 levels will be tracked directly to see how clean the air is. The policy will be tested based on how it keeps CO2 emissions low and stops CO2 levels from going above safe limits when the building is occupied.

Finally the evaluation will measure actuator stability by analyzing the smoothness of the action signals produced by the agent. I think actuator stability is important. It is ensured because the reinforcement learning policy does not generate erratic high-frequency command changes that could physically degrade the real-world dampers and valves over time.

12. Expected Novelty

Instead Of giving simple separate temperature setpoints such as "set AC to 22°C" the model-free policy creates smooth continuous action signals. This lets the AI talk directly to the building's parts—adjusting dampers, variable-speed fans and chilled-water valves, in the same way a real building management system would.

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